Project Report: Diwali Sales Exploratory Data Analysis

1. Project Overview

The objective of this EDA is to analyze a Diwali sales dataset to uncover demographic purchasing patterns, evaluate regional and occupational sales performance, and provide actionable, data-driven recommendations to optimize future targeted marketing campaigns and inventory planning.

2. Methodology & Steps Involved

- **Data Cleaning:** Standardized column names, dropped irrelevant columns (status, unnamed1), and removed rows with missing amount values to ensure mathematical accuracy.
- **Feature Engineering:** Converted numeric binary codes in marital_status to readable text ("Un-married", "Married"). Applied **Median Target Encoding** to categorical variables to enable correlation matrix calculations.
- **Exploratory Data Analysis (EDA):** Leveraged Pandas groupby and agg functions to calculate descriptive statistics (sums, medians, counts) across demographics and geographic zones.
- **Data Visualization:** Designed multivariate visualizations using Seaborn and Matplotlib, including boxplots to show dispersion/outliers, and dual-axis combo charts to compare total revenue against order volume simultaneously.

3. Key Business Questions Answered

- Which demographic segment (age, gender, marital status) drives the most overall revenue?
- Do different demographic groups have different individual purchasing power (median spend)?
- Which occupational sectors and geographic states bring in the highest volume of orders vs. the highest value per order?
- What product categories are the primary drivers of holiday revenue?

4. Key Insights

- **The Power of Volume vs. Value:** While the IT Sector, Healthcare, and Aviation dominate total overall sales, the *Chemical* sector yields the highest average amount per individual order (4,046).
- **Demographic Dominance:** Un-married females are the core customer base, driving nearly 4.38 Crore in sales. However, median individual spending remains highly consistent (around 8,000) across all genders and marital statuses. The revenue gap is driven entirely by the number of transactions, not individual purchasing power.
- **Age and Outliers:** The 26-35 age group is the most lucrative segment (4.26 Crore). Boxplot distributions reveal that while baseline spending is uniform, high-ticket outlier purchases are primarily driven by younger and middle-aged demographics.
- **Category Leaders:** Food (3.39 Crore), Clothing (1.64 Crore), and Electronics (1.56 Crore) lead total sales. Geographically, Uttar Pradesh, Maharashtra, and Karnataka are the highest-performing states.
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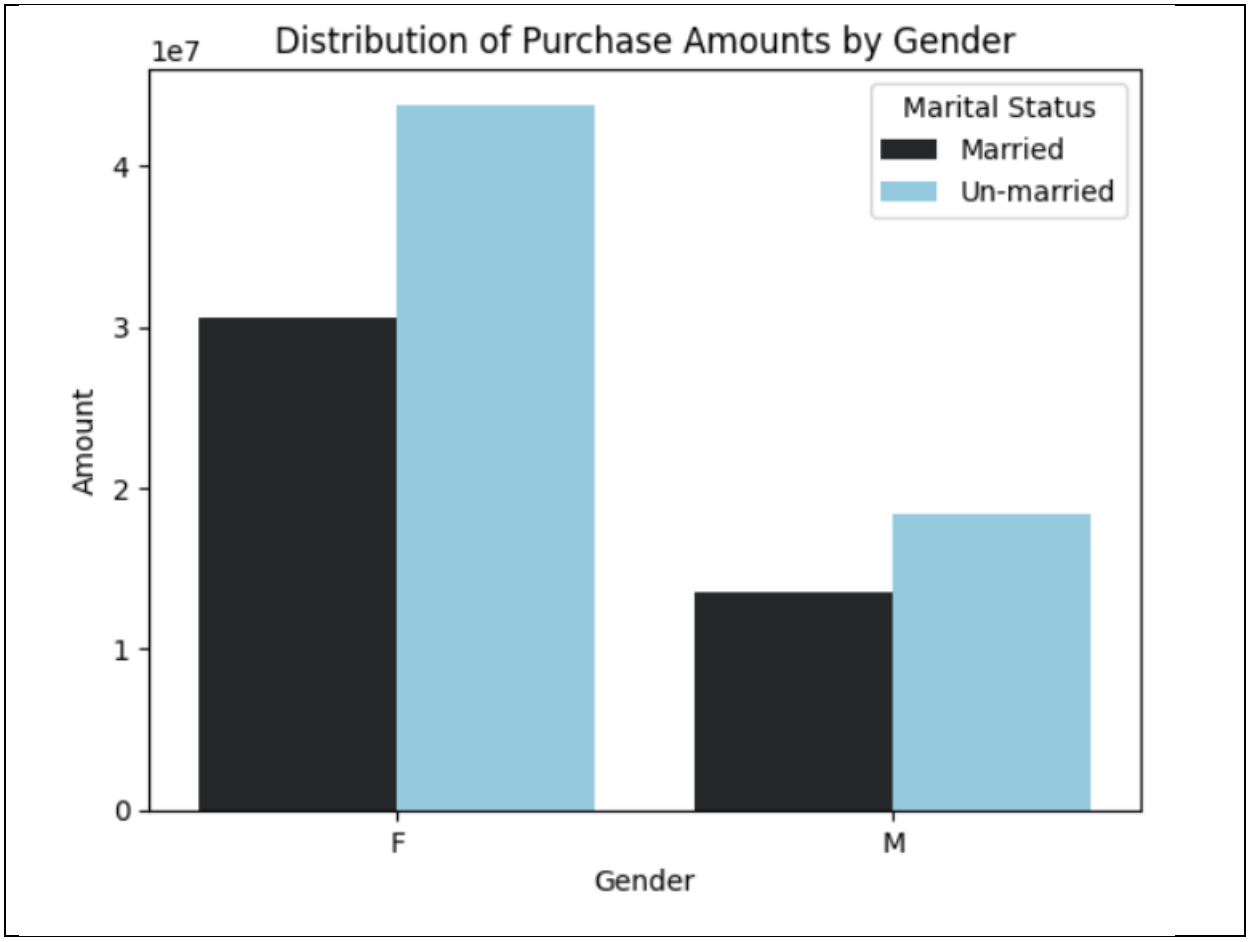
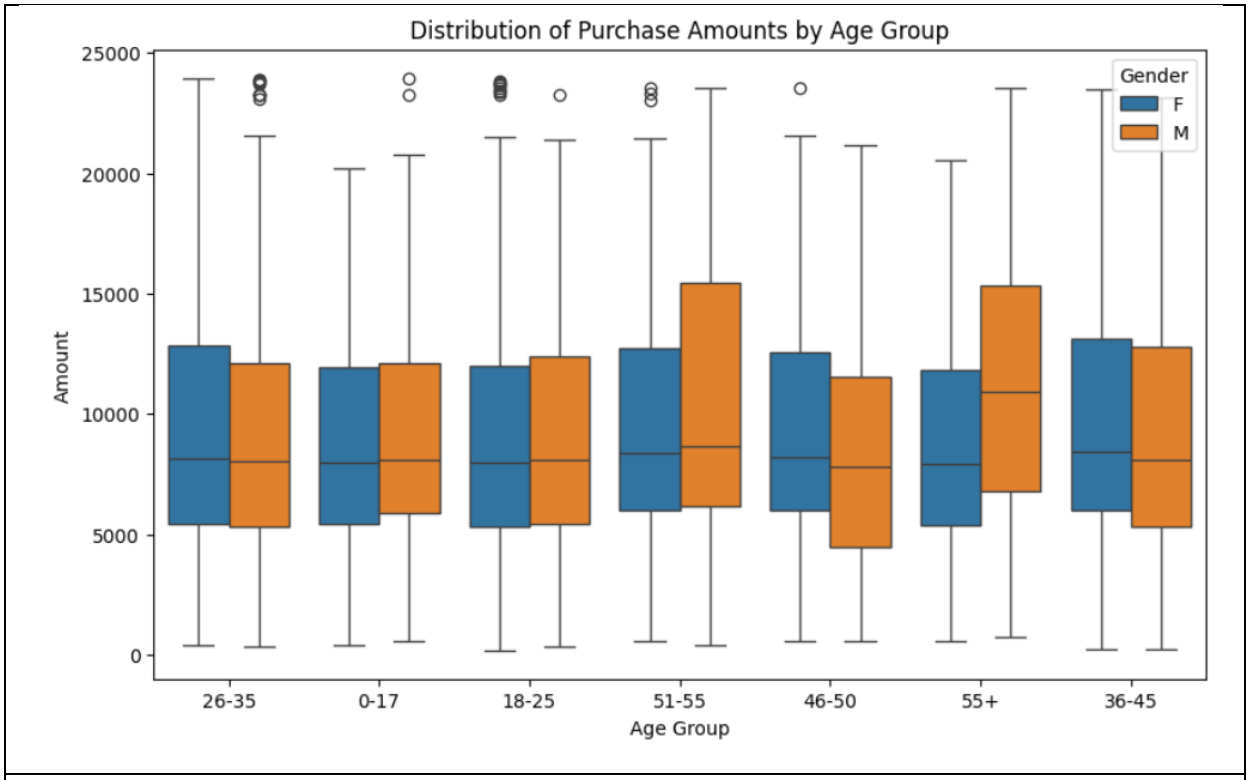
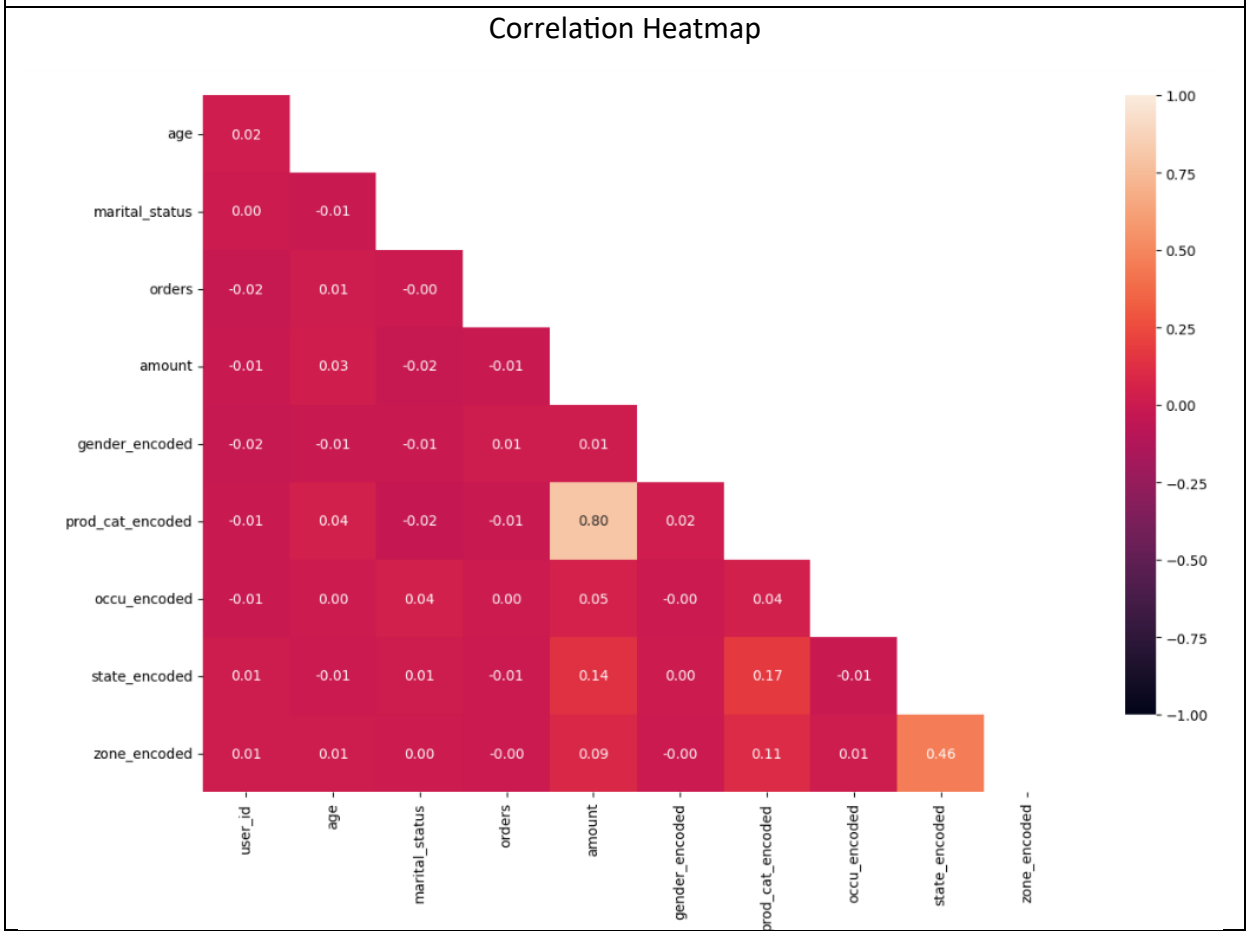
5. Business Recommendations

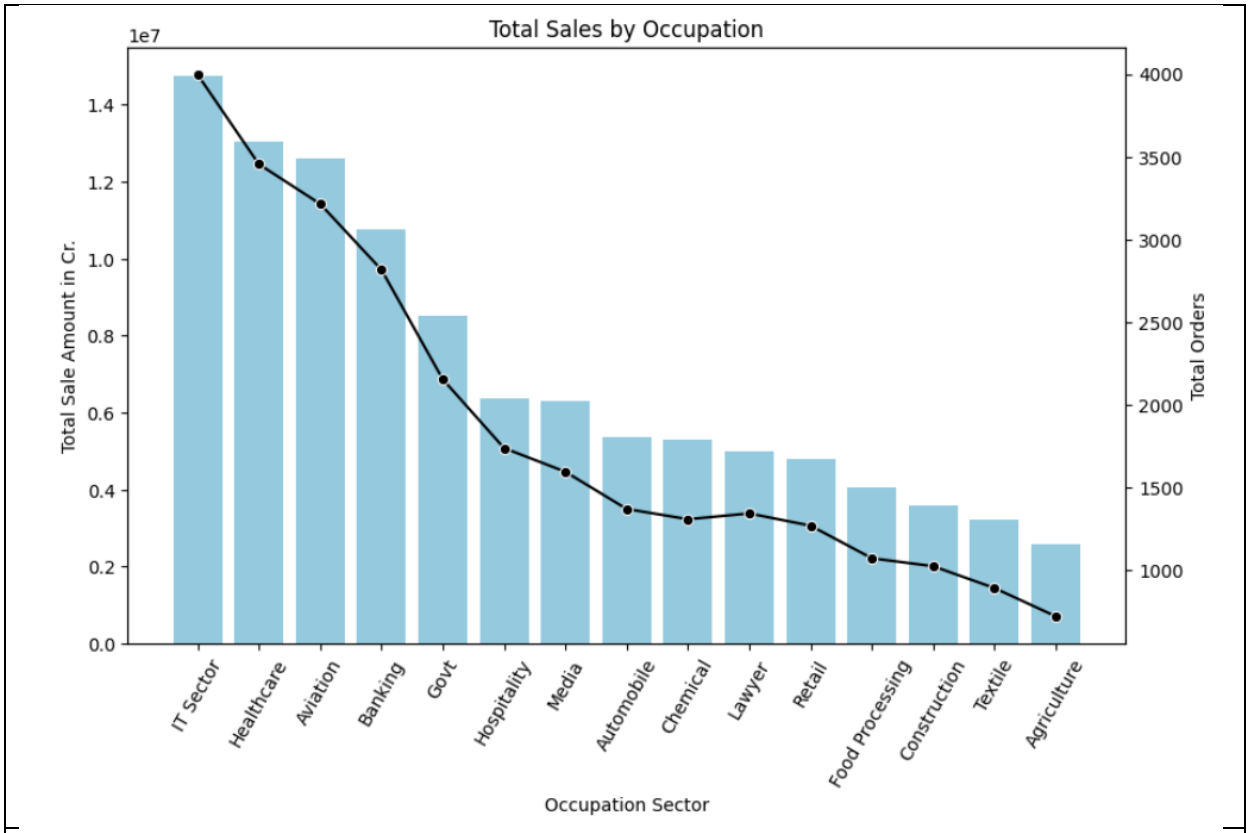
- **Targeted Ad Spend:** Marketing budgets for the next holiday season should be aggressively reallocated to target un-married women aged 26-35 working in IT, Healthcare, and Aviation.
- **Premium Upselling:** Since the Chemical sector shows the highest average order value, create targeted "premium" or bundle-deal campaigns for customers in this sector to maximize their high purchasing power.
- **Inventory Focus:** Ensure supply chain readiness for Food, Clothing, and Electronics in warehouses servicing Uttar Pradesh and Maharashtra, as these combinations represent the highest sheer volume of orders

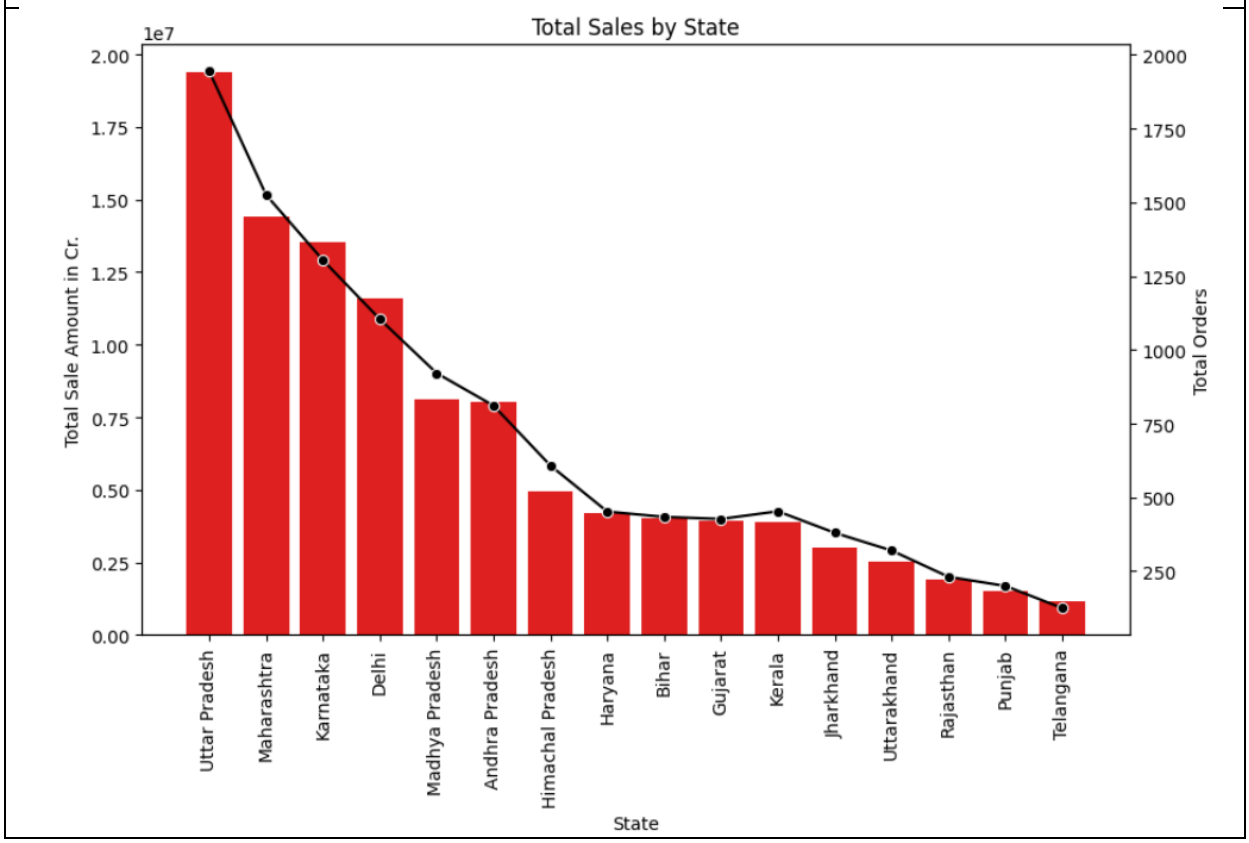
6. Graphs

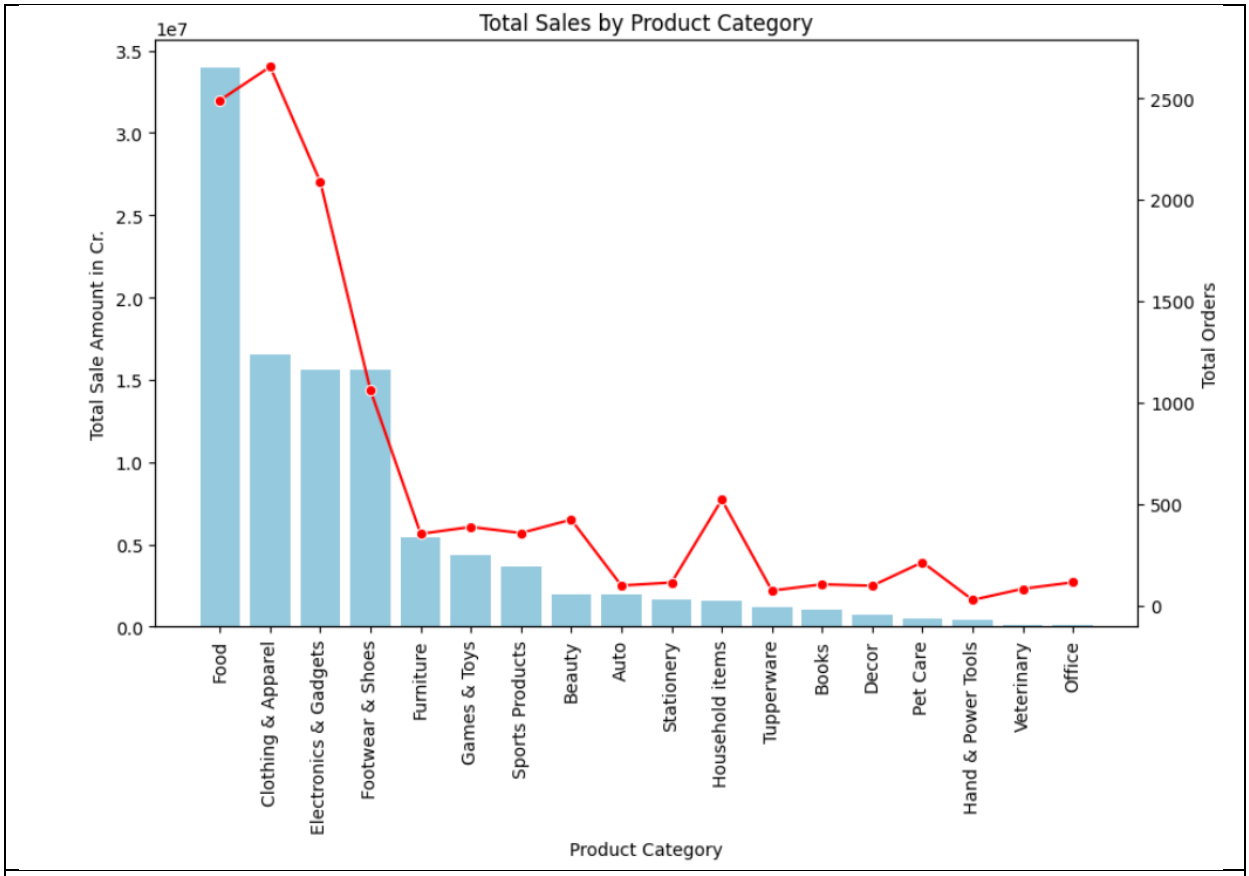


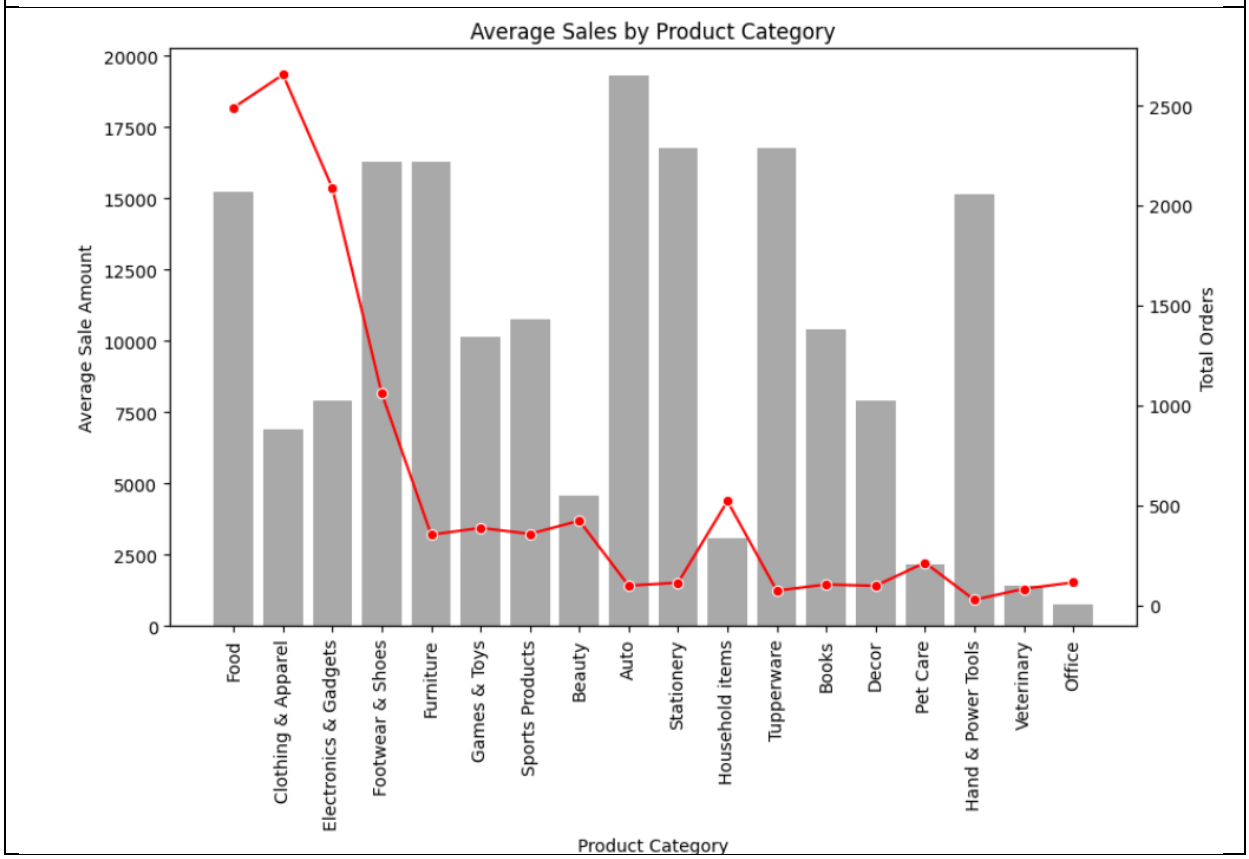

Distribution of Purchase Amounts by Age Group
Amount
Age Group
Gender
F
M
25000
20000
15000
10000
5000
0
26-35
0-17
18-25
51-55
46-50
55+
36-45


Correlation Heatmap
age
marital_status
orders
amount
gender_encoded
prod_cat_encoded
occu_encoded
state_encoded
zone_encoded
user_id
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00


Total Sales by Occupation
1e7
Total Sale Amount in Cr.
Total Orders
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
1000
1500
2000
2500
3000
3500
4000
IT Sector
Healthcare
Aviation
Banking
Govt
Hospitality
Media
Automobile
Chemical
Lawyer
Retail
Food Processing
Construction
Textile
Agriculture
Occupation Sector


Total Sales by State
1e7
Total Sale Amount in Cr.
Total Orders
0.00
0.25
0.50
0.75
1.00
1.25
1.50
1.75
2.00
250
500
750
1000
1250
1500
1750
2000
Uttar Pradesh
Maharashtra
Karnataka
Delhi
Madhya Pradesh
Andhra Pradesh
Himachal Pradesh
Haryana
Bihar
Gujarat
Kerala
Jharkhand
Uttarakhand
Rajasthan
Punjab
Telangana
State


Total Sales by Product Category
1e7
Total Sale Amount in Cr.
Total Orders
Product Category
Food
Clothing & Apparel
Electronics & Gadgets
Footwear & Shoes
Furniture
Games & Toys
Sports Products
Beauty
Auto
Stationery
Household items
Tupperware
Books
Decor
Pet Care
Hand & Power Tools
Veterinary
Office


Average Sales by Product Category
Average Sale Amount
Total Orders
Product Category
Food
Clothing & Apparel
Electronics & Gadgets
Footwear & Shoes
Furniture
Games & Toys
Sports Products
Beauty
Auto
Stationery
Household items
Tupperware
Books
Decor
Pet Care
Hand & Power Tools
Veterinary
Office